Project V: Anomaly Detection in Manufacturing

Scenario

Your team is part of an analytics department at a large manufacturer producing printed circuit boards. Throughout the different production stages, the company has noticed a recent rise in machine breakdowns. As a consequence, to avoid preventable downtime in the production facility, leading to high added costs, a predictive maintenance setup should be implemented.

Therefore, you are tasked with developing an ML-based anomaly detection system that accurately captures machine breakdowns, providing operations staff with an indication of machines requiring maintenance. The organization is particularly interested in the main factors that signify machine wear. For this purpose, you are equipped with a labeled data set of production schedules, showing the occurrence of machine breakdowns. For guiding the implementation, adopt the introduced MLOps-based process model to ensure transparency in your development, leading to an effective and maintainable system.

Data

Attribute	Description
ID	Identifier for a job within a schedule
Priority	Priority/deadline of the job
Family_type	Family type of a job
First_stage	ID of the machine used to process the job in the first stage (see Fig. 1)
Start_time_S1l	Start time of job on first processing stage
Finish_time_S1	End time of job on first processing stage
Processing_Time_S1	Processing time of job on first stage
Second_stage	ID of the machine used to process the job in the second stage (see Fig. 1)
Start_time_S2	Start time of job on second processing stage
Finish_Time_S2	End time of job on second processing stage
Processing_Time_S2	Processing time of job on second stage
Third_stage	ID of the machine used to process the job in the third stage (see Fig. 1)
Start_time_S3	Start time of job on third processing stage
Finish_time_S3	End time of job on third processing stage
Processing_Time_S3	Processing time of job on third stage
Fourth_stage	ID of the machine used to process the job in the fourth stage (see Fig. 1)
Start_time_s4	Start time of job on fourth processing stage
Finish_time	End time of job
Processing_Time_S4	Processing time of job on fourth stage
Overall_processing_time	Total processing time of job
Overall_waiting_time	Total waiting time of job in the production system
Tardiness	Tardiness of a job
BREAKS	Number of machine breakdowns for a job

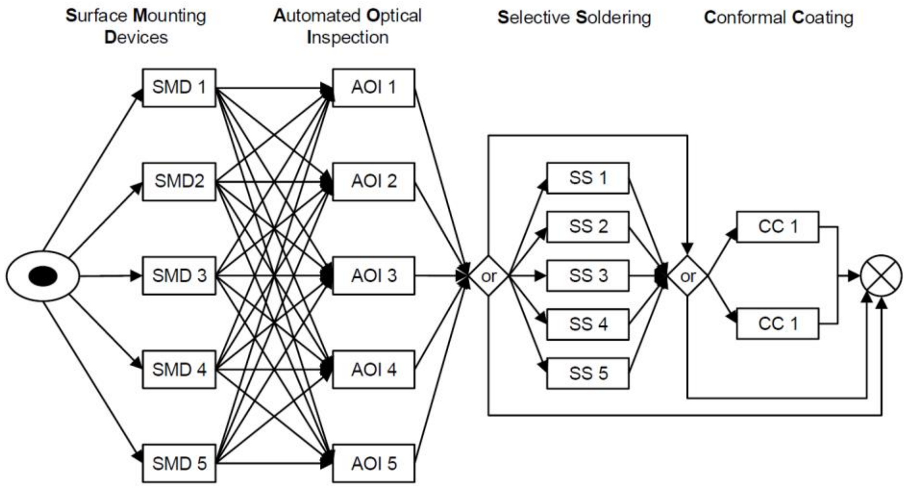

Figure 1. Production environment

Deliverables

- Extensive report containing:
    - Detailed documentation of the implementations in accordance with the steps and components of the used MLOps-based process model
    - Applied patterns and concepts from the lecture and exercise
    - Explanation of design decisions
    - Outputs in the context of the business scenario
- Code base
- Additionally used data (optional)